

$$\textcircled{1} a) N \times N = N^2 \times 2 \text{ FLOPs} = \boxed{2N^2} = 2(512)^2 = 524288 \text{ FLOPs}$$

$$b) N^2 \text{ elements} \times 4 \text{ bytes} = \boxed{4N^2} = 4(512)^2 = 1048576 \text{ bytes}$$

$$c) 2N^2 \text{ FLOPs} \times (1-s) \text{ fraction of zeros} = \boxed{2N^2(1-s)}$$

$$d) N^2(1-s) \times 4 \text{ bytes} + N^2(1-s) \times 4 \text{ bytes} = 8N^2(1-s)$$

$$\textcircled{2} 2 = \frac{2N^2}{2N^2(1-s)} = \boxed{8N^2(1-s) + 4(N+1)}$$

$$2 = \frac{1}{(1-s)}$$

$$2(1-s) = 1$$

$$2 - 2s = 1$$

$$-2s = -1$$

$$\boxed{s = 1/2}$$

$$\textcircled{3} 8N^2(1-s) + 4(N+1) = 4N^2$$

$$8N^2 - 8N^2s + 4N + 4 = 4N^2$$

$$-8N^2s + 4N + 4 = -4N^2$$

$$2N^2s - N - 1 = N^2$$

$$2N^2s = N^2 + N + 1$$

$$\boxed{s = \frac{N^2 + N + 1}{2N^2} = \frac{(512)^2 + (512) + 1}{2(512)^2} = 0.50047 \approx 0.501}$$

$$\textcircled{4} \text{ Dense} = 4(512)^2 = 1048576 \text{ bytes}$$

$$\text{Sparse} = 8(512)^2(1-0.9) + 4(512+1) = 211767.2 \text{ bytes}$$

$$\frac{1048576 \text{ bytes}}{211767.2 \text{ bytes}} \approx \boxed{4.95x}$$